

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Ist Semester of BSc (Biological sciences) Examination March, 2019

BE112 Biotechnology and Human Welfare

Date: 19/03/2019 Day: Tuesday Time: 10.00 a.m. To 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions. **20**

1. In a cell, mitochondrial DNA is present in high copy number.

☐ True

☐ False

2. _____ is the first step in any biotechnological process.

- (i) Identification of organism or cell type
- (ii) Cell response optimization
- (iii) Mass culture
- (iv) None of the above

3. Application of *Bacillus thuringiensis* in crops destroys

- (i) Rodents (ii) Fungi (iii) Insects (iv) None of these

4. PEG used in protoplast fusion stands for

- (i) polyethane glyoxalic acid (ii) polyethylene glycol
- (iii) propionate ethylene glycol (iv) none of these

5. One of the following type of bonds is broken down by Amylase enzyme
(i) glycosidic (ii) peptide (iii) ester (iv) phosphodiester
6. Main component of soil humus is/are
(i) Actinomycetes (ii) Humic acid
(iii) fulvic acid (iv) both ii & iii
7. PSMs are a group of plant growth promoting microorganisms that carry solubilization of insoluble phosphate.

☐True ☐False
8. *Saccharomyces cerevisiae* is
(i) yeast (ii) bacteria (iii) Algae (iv) fungus
9. Biotechnological applications involving development in agriculture sector is ____
(i) Green biotechnology (ii) Red biotechnology
(iii) Blue biotechnology (iv) White biotechnology
10. One of the following is not related to 'detergent application'
(i) Amylases (ii) Lipases (iii) Proteases (iv) Keratinases
11. *Nitrogenase* enzyme found in heterocyst of Cyanobacteria helps in fixation of
(i) P (ii) N (iii) K (iv) none of these
12. Polymerase chain reaction technique utilizes the application of enzyme Taq Polymerase.

☐True ☐False
13. The method of DNA finger printing was developed by _____.
(i) Rosalind Franklin (ii) Alec Jeffreys
(iii) Har Gobind Khorana (iv) James Watson
14. Bioremediation is the application of microbes for removal of heavy metal from heavy metal contaminated sites. This application is
(i) medical biotechnology (ii) industrial biotechnology
(iii) forensic (iv) none of these

15. Mycorrhiza is responsible for _____ the surface area for soil water & nutrients absorption by plant roots.
i) increasing ii) decreasing iii) no change iv) none of these
16. Antibodies found in our body are complex
(i) Lipoproteins (ii) Prostaglandins (iii) Glycoproteins (iv) Steroids
17. Insulin is a
(i) carbohydrate (ii) hormone (iii) lipid (iv) none of these
18. On the planet earth the DNA of every human being is 99.9% same.
☐True ☐False
19. Which of the following type of molecule is utilised by PCR for diagnostic purpose?
(i) Nucleic acids (ii) Proteins (iii) Lipids (iv) Carbohydrates
20. DBT stands for Department of Biotechnology.
☐True ☐False

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BE112 Biotechnology and Human Welfare

Date: 19/03/2019 Day: Tuesday Time: 10.30 a.m. To 1.00 p.m. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed

20

Answer any four of the following:

- (1) Differentiate liquefaction and saccharification processes for alcohol formation. Write down the applications of proteases and amylases.
- (2) Define Genetically Modified Foods. What are the benefits of flavor savor GM tomato? How do Bt endotoxins controls insect attack in GM crops?
- (3) Briefly explain the 'protoplast fusion' and 'biolistic method' of gene transformation.
- (4) What is a vaccine? Give few examples. Write a brief note on DNA vaccine.
- (5) Define DNA fingerprinting. Describe several applications of DNA fingerprinting.
- (6) Diagrammatically explain the process of nodule formation in a legume root.

SECTION - II

Q-III Answer the following questions as directed

30

Answer any six of the following:

- (1) Describe the biological nitrogen fixation (BNF) cycle. Discuss the role and importance of microorganisms in this process.
- (2) Write down the steps involved in any biotechnological process. Describe the applications of Biotechnology in Dairy industry.
- (3) Give a detailed note on the applications of Gene Therapy.
- (4) What are biodegradable polymers? Discuss the manufacturing and applications of biodegradable plastics.
- (5) Define monoclonal antibodies and discuss their production method.
- (6) Define livestock. Explain in detail which traits of livestock are improved using biotechnological methods?
- (7) Discuss the recovery and purification process of enzymes.
- (8) Name any four biological samples used in forensics. Write a note on RAPD method in detail.